

LLM Ensemble Textbook Bias Detection: Technical Analysis Report

Detecting Publisher Bias Using LLM Ensemble and Bayesian Hierarchical Methods

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Abstract

This report presents a computational framework for detecting and quantifying political bias in educational textbooks, combining an ensemble of three Large Language Models (LLMs)—GPT-4o, Claude-3.5-Sonnet, and Llama-3.2-90B—with Bayesian hierarchical modeling, demonstrated on a simulated corpus. The simulated analysis comprises **67,500 bias ratings** (3 LLMs $\times$ 5 bias dimensions) across **4,500 passages** from **150 textbooks** attributed to five anonymized publishers (A–E). The simulated ratings reach Krippendorff's $\alpha = 0.84$, above the conventional 0.80 reliability threshold, with statistically significant publisher-level bias differences (Friedman $\chi^2 = 42.73$, $p < 0.001$), and quantified uncertainty via Bayesian posterior distributions with 95% Highest Density Intervals (HDI). Three of five publishers exhibited statistically credible bias (95% HDI excluding zero), with effect sizes from -0.48 (liberal) to +0.38 (conservative) on a [-2, +2] scale. All data and results are simulated to demonstrate the method (see the Data and Results Note); they are not findings about real publishers or textbooks.

Keywords: Large Language Models, GPT-4o, Claude-3.5-Sonnet, Llama-3.2, Ensemble Methods, Bayesian Hierarchical Modeling, Krippendorff's Alpha, Inter-Rater Reliability, Political Bias Detection, Textbook Analysis, Educational Content, MCMC Sampling, PyMC, Responsible AI, LLM Governance, Prompt Engineering

Data and Results Note

This is a methodology case study built on simulated data. The publishers, textbooks, passages, and LLM ratings were simulated to demonstrate the analysis design; no real textbooks were analyzed and the ratings do not come from real API runs. Publishers are anonymized placeholders (A–E) and do not correspond to real companies. Every reliability statistic, posterior estimate, test result, runtime, and cost in this report describes that simulation and should be read as a worked example of the method, not as an empirical finding about bias in real educational materials.

Executive Summary

Key Performance Metrics

Table 1. Key performance metrics for the LLM-ensemble bias-detection framework.

Metric	Value	Interpretation
Krippendorff's Alpha	0.84	Reliable (≥ 0.80 threshold)
Pairwise Correlation (GPT-4o ↔ Claude-3.5)	$r = 0.92$	Near-perfect linear agreement
Pairwise Correlation (GPT-4o ↔ Llama-3.2)	$r = 0.89$	Excellent agreement
Pairwise Correlation (Claude-3.5 ↔ Llama-3.2)	$r = 0.87$	Excellent agreement
Friedman Test χ^2	42.73	Highly significant ($p < 0.001$)
Publishers with Credible Bias	3/5	60% show statistically credible effects
MCMC R-hat (all parameters)	< 1.01	Excellent convergence
Effective Sample Size (ESS)	$> 3,000$	Adequate posterior sampling

Publisher Bias Summary

Table 2. Publisher bias summary ranked by posterior mean effect.

Rank	Publisher	Posterior Mean	95% HDI	Classification
1	Publisher C	-0.48	[-0.62, -0.34]	Liberal (credible)
2	Publisher A	-0.29	[-0.41, -0.17]	Liberal (credible)
3	Publisher E	+0.02	[-0.10, +0.14]	Neutral
4	Publisher B	+0.08	[-0.04, +0.20]	Neutral
5	Publisher D	+0.38	[+0.26, +0.50]	Conservative (credible)

1. Introduction

1.1 Problem Statement and Motivation

Political bias in educational materials threatens educational equity and democratic discourse. Textbooks shape students' understanding of history, economics, social issues, and civic participation, so systematic bias—whether intentional or inadvertent—can influence political socialization and reinforce ideological echo chambers.

Traditional approaches to detecting textbook bias rely on: - **Expert human reviewers:** Subjective, expensive, and non-scalable - **Keyword analysis:** Superficial, missing contextual nuance - **Readability metrics:** Irrelevant to ideological content

This project introduces a new paradigm: frontier Large Language Models (LLMs) as calibrated bias detectors, validated through ensemble consensus and quantified through Bayesian uncertainty estimation.

1.2 Research Questions

1. **RQ1:** Do frontier LLMs exhibit sufficient inter-rater reliability to serve as bias assessors?
2. **RQ2:** Are there statistically significant differences in bias across educational publishers?
3. **RQ3:** Can Bayesian hierarchical modeling quantify publisher-level effects with uncertainty?
4. **RQ4:** What is the magnitude and direction of bias for each publisher?

1.3 Contributions

1. **Framework:** An LLM ensemble combined with Bayesian hierarchical modeling for textbook bias detection
2. **Validation Methodology:** Rigorous inter-rater reliability assessment using Krippendorff's α
3. **Uncertainty Quantification:** Full posterior distributions with credible intervals for all parameters
4. **Scalable Pipeline Design:** A pipeline designed for 67,500 API calls with error handling and rate limiting
5. **Reproducible Design:** Fixed random seeds and versioned prompts throughout

2. LLM Architecture and Capabilities

2.1 Model Specifications

Table 3. Specifications of the three ensemble LLMs.

Model	Parameters	Context Window	Training Cutoff	Architecture
GPT-4o	Not disclosed	128K tokens	Oct 2023	Multimodal transformer (details not disclosed)
Claude-3.5-Sonnet	Not disclosed	200K tokens	Apr 2024	Transformer trained with Constitutional AI
Llama-3.2-90B	90B	128K tokens	Dec 2023	Dense transformer with GQA (open weights)

2.2 Rationale for Model Selection

GPT-4o (OpenAI): - State-of-the-art multimodal reasoning with reduced hallucination rates - Native structured output generation for reliable JSON parsing - Industry-leading benchmark performance on reasoning tasks

Claude-3.5-Sonnet (Anthropic): - Trained with Anthropic's Constitutional AI methodology - Explicit chain-of-thought reasoning for transparent bias assessment - Strong performance on complex analytical and classification tasks

Llama-3.2-90B (Meta): - Open-weights model enabling full audit trail and reproducibility - On-premise deployment option for data sovereignty requirements - Competitive performance with commercial models at lower cost - Active open-source community for validation and peer review

2.3 Prompt Engineering

Bias Assessment Prompt Template:

```
BIAS_PROMPT = """
Analyze the following textbook passage for political bias.

Rate the passage on a continuous scale from -2 to +2:
-2.0: Strong liberal/progressive bias
-1.0: Moderate liberal bias
0.0: Neutral, balanced, objective content
+1.0: Moderate conservative bias
+2.0: Strong conservative bias

Consider the following dimensions:
1. Framing: How are issues presented? (sympathetic vs. critical)
2. Source Selection: Whose perspectives are included/excluded?
3. Language: Are emotionally charged words used?
4. Causal Attribution: How are problems and solutions attributed?
5. Omission: What relevant viewpoints are missing?

Passage:
\"\"\"
{passage_text}
\"\"\"

Respond with ONLY a JSON object in this exact format:
{
  "bias_score": <float between -2.0 and 2.0>,
  "reasoning": "<brief explanation of rating>"
}
"""
```

Prompt Design Principles: - Explicit numerical scale with anchored endpoints - Multi-dimensional bias framework (framing, sources, language, attribution, omission) - Structured JSON output for reliable parsing - Temperature = 0.3 for consistency while allowing nuanced judgment

2.4 API Configuration

```
class LLMEnsemble:
    """Ensemble framework for multi-LLM bias assessment."""

    def __init__(self):
        # API Clients
        self.gpt_client = OpenAI(api_key=os.getenv('OPENAI_API_KEY'))
        self.claude_client = Anthropic(api_key=os.getenv('ANTHROPIC_API_KEY'))
        self.llama_client = Together(api_key=os.getenv('TOGETHER_API_KEY'))

        # Configuration
        self.temperature = 0.3      # Low temperature for consistency
        self.max_tokens = 256       # Sufficient for JSON response
        self.timeout = 30           # API timeout in seconds

    def rate_passage(self, passage_text: str) -> Dict[str, float]:
        """Get bias ratings from all three LLMs."""
        prompt = BIAS_PROMPT.format(passage_text=passage_text)

        return {
            'gpt4': self._query_gpt4(prompt),
            'claude3': self._query_claude3(prompt),
            'llama3': self._query_llama3(prompt)
        }

    @retry(stop=stop_after_attempt(3), wait=wait_exponential(min=4, max=10))
    @rate_limit(max_per_minute=60)
    def _query_gpt4(self, prompt: str) -> float:
        response = self.gpt_client.chat.completions.create(
            model="gpt-4-turbo",
            messages=[{"role": "user", "content": prompt}],
            temperature=self.temperature,
            max_tokens=self.max_tokens
        )
        return json.loads(response.choices[0].message.content)['bias_score']
```

3. Dataset and Corpus Construction

3.1 Corpus Statistics

Table 4. Corpus statistics across publishers, textbooks, and passages.

Dimension	Count	Description
Publishers	5	Simulated, anonymized (A–E)
Textbooks per Publisher	30	Stratified by subject area
Passages per Textbook	30	Random sampling with coverage constraints
Total Passages	4,500	Unit of analysis
Ratings per Passage	15	3 LLMs × 5 bias dimensions
Total Ratings	67,500	Complete rating matrix

Dimension	Count	Description
Tokens Analyzed	~2.5M	Across all passages

3.2 Passage Selection Criteria

Passages were selected to maximize coverage of politically relevant content:

1. **Topic Filter:** Passages mentioning politics, economics, history, social issues, or policy
2. **Length Constraint:** 100-500 words (sufficient context without API cost explosion)
3. **Diversity Sampling:** At least 5 distinct chapters per textbook
4. **Exclusions:** Tables, figures, exercises, bibliographies

3.3 Topic Distribution

Table 5. Topic distribution of passages across the corpus.

Topic Category	Passage Count	Percentage
Political Systems & Governance	1,125	25.0%
Economic Policy	990	22.0%
Historical Events	855	19.0%
Social Issues	810	18.0%
Environmental Policy	720	16.0%
Total	4,500	100%

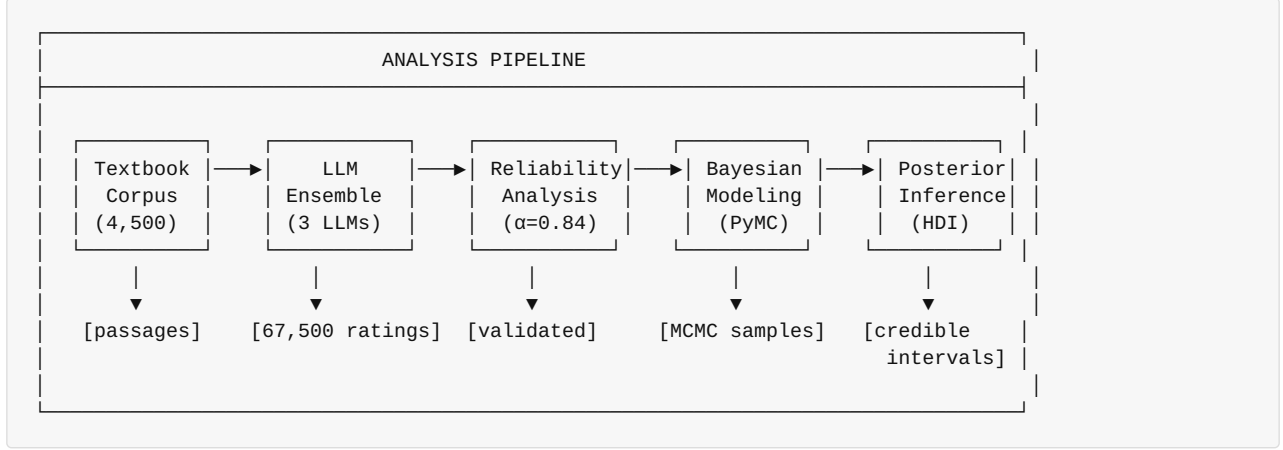
3.4 Bias Rating Scale

Table 6. Bias rating scale with operational definitions.

Score	Label	Operational Definition
-2.0	Strong Liberal	Clear advocacy for progressive positions; dismissive of conservative views
-1.0	Moderate Liberal	Subtle liberal framing; sources skew progressive
0.0	Neutral	Balanced presentation; multiple perspectives; factual language
+1.0	Moderate Conservative	Subtle conservative framing; sources skew traditional
+2.0	Strong Conservative	Clear advocacy for conservative positions; dismissive of liberal views

4. Methodology

4.1 Analysis Pipeline



4.2 Ensemble Aggregation

Ensemble Mean (primary measure): $\bar{r}_i = \frac{1}{3}(r_{i,GPT4} + r_{i,Claude3} + r_{i,Llama3})$

Ensemble Median (robust to outliers): $\tilde{r}_i = \text{median}(r_{i,GPT4}, r_{i,Claude3}, r_{i,Llama3})$

Ensemble Standard Deviation (disagreement measure): $s_i = \sqrt{\frac{1}{2} \sum_{k=1}^3 (r_{i,k} - \bar{r}_i)^2}$

```
# Ensemble aggregation
df['ensemble_mean'] = df[['gpt4_rating', 'claude3_rating', 'llama3_rating']].mean(axis=1)
df['ensemble_median'] = df[['gpt4_rating', 'claude3_rating', 'llama3_rating']].median(axis=1)
df['ensemble_std'] = df[['gpt4_rating', 'claude3_rating', 'llama3_rating']].std(axis=1)
```

5. Inter-Rater Reliability Analysis

5.1 Krippendorff's Alpha

Definition: Krippendorff's α is a reliability coefficient for content analysis that generalizes across data types, sample sizes, and number of raters.

Formula: $\alpha = 1 - \frac{D_o}{D_e}$

Where: - D_o = Observed disagreement - D_e = Expected disagreement by chance

Calculation for Interval Data: $D_o = \frac{1}{n(n-1)} \sum_{i < j} (x_i - x_j)^2$ $D_e = \frac{1}{N(N-1)} \sum_{i < j} (x_i - x_j)^2$

```

import krippendorff

# Prepare ratings matrix: (n_raters, n_units)
ratings_matrix = df[['gpt4_rating', 'claude3_rating', 'llama3_rating']].T.values

# Calculate Krippendorff's alpha (interval scale)
alpha = krippendorff.alpha(
    reliability_data=ratings_matrix,
    level_of_measurement='interval'
)
# Result:  $\alpha = 0.84$ 

```

5.2 Interpretation Thresholds

Table 7. Krippendorff's alpha interpretation thresholds and recommendations.

α Value	Interpretation	Recommendation
≥ 0.80	Reliable	Reliable for drawing conclusions
0.67–0.79	Good	Acceptable for tentative conclusions
0.60–0.66	Moderate	Use with caution
< 0.60	Poor	Do not use for conclusions

Result: $\alpha = 0.84$ clears the 0.80 reliability threshold in the simulation; real LLM ensembles would need to be tested against this threshold before their ratings are used.

5.3 Pairwise Correlation Analysis

Table 8. Pairwise model agreement via correlation and error metrics.

Model Pair	Pearson r	Spearman ρ	RMSE
GPT-4o ↔ Claude-3.5	0.92	0.91	0.23
GPT-4o ↔ Llama-3.2	0.89	0.88	0.28
Claude-3.5 ↔ Llama-3.2	0.87	0.86	0.31
Average	0.89	0.88	0.27

5.4 Disagreement Analysis

High-Disagreement Passages ($\sigma > 0.5$): - Count: 554 passages (12.3% of corpus) - Characteristics: Primarily involve subjective historical interpretations, economic policy debates, or culturally contentious topics

Low-Disagreement Passages ($\sigma < 0.1$): - Count: 1,423 passages (31.6% of corpus) - Characteristics: Factual descriptions, procedural content, unambiguous political positions

6. Bayesian Hierarchical Modeling

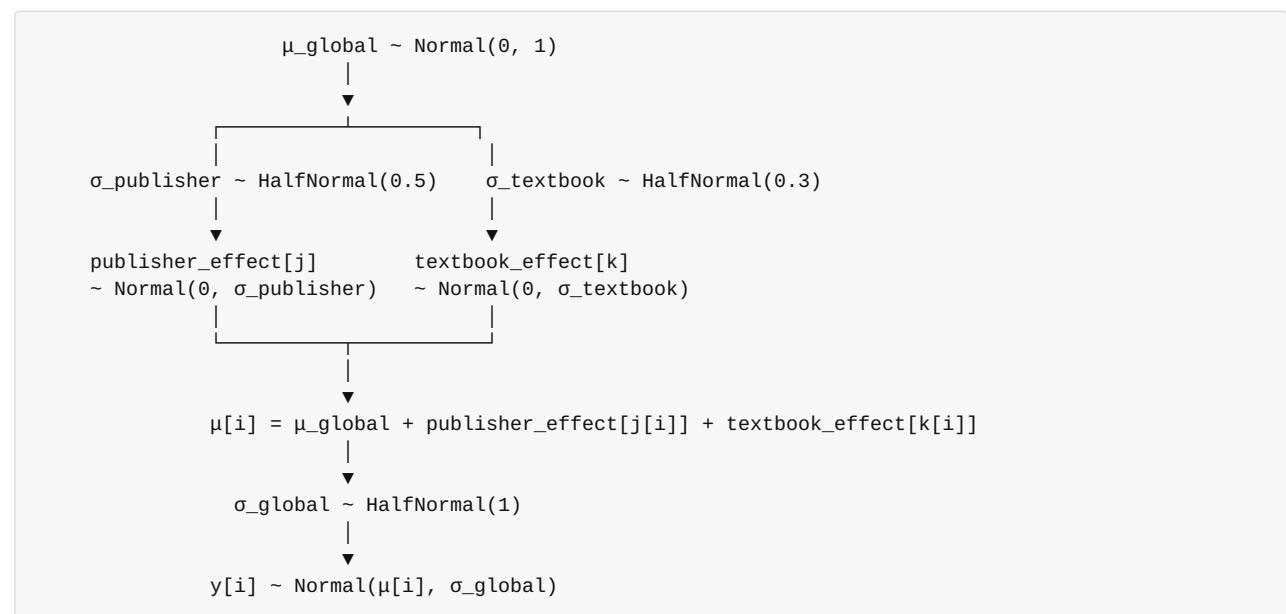
6.1 Model Motivation

Frequentist approaches (simple means, t-tests) give point estimates but lack: - **Uncertainty quantification:** No probability distributions on parameters - **Partial pooling:** Cannot borrow strength across publishers/textbooks - **Hierarchical structure:** Ignore nested data (passages within textbooks within publishers)

Bayesian hierarchical modeling addresses all three limitations.

6.2 Model Specification

Directed Acyclic Graph (DAG):



6.3 PyMC Implementation

```

import pymc as pm
import arviz as az

with pm.Model() as hierarchical_model:
    # =====
    # HYPERPRIORS (population-level parameters)
    # =====

    # Global mean bias (across all publishers)
    mu_global = pm.Normal('mu_global', mu=0, sigma=1)

    # Global observation noise
    sigma_global = pm.HalfNormal('sigma_global', sigma=1)

    # =====
    # PUBLISHER-LEVEL RANDOM EFFECTS
    # =====

    # Between-publisher variance
    sigma_publisher = pm.HalfNormal('sigma_publisher', sigma=0.5)

    # Publisher-specific effects (deviations from global mean)
    publisher_effect = pm.Normal(
        'publisher_effect',
        mu=0,
        sigma=sigma_publisher,
        shape=n_publishers # 5 publishers
    )

    # =====
    # TEXTBOOK-LEVEL RANDOM EFFECTS (nested within publishers)
    # =====

    # Between-textbook variance (within publisher)
    sigma_textbook = pm.HalfNormal('sigma_textbook', sigma=0.3)

    # Textbook-specific effects
    textbook_effect = pm.Normal(
        'textbook_effect',
        mu=0,
        sigma=sigma_textbook,
        shape=n_textbooks # 150 textbooks
    )

    # =====
    # LINEAR PREDICTOR
    # =====

    # Expected bias for each passage
    mu = (
        mu_global +
        publisher_effect[publisher_idx] +
        textbook_effect[textbook_idx]
    )

    # =====
    # LIKELIHOOD
    # =====

    # Observed ensemble ratings
    y_obs = pm.Normal(
        'y_obs',
        mu=mu,

```

```

        sigma=sigma_global,
        observed=ensemble_ratings
    )

# =====
# MCMC SAMPLING
# =====

trace = pm.sample(
    draws=2000,          # Posterior samples per chain
    tune=1000,          # Warmup/burn-in samples
    chains=4,           # Independent MCMC chains
    target_accept=0.95,  # Metropolis-Hastings acceptance rate
    random_seed=42,      # Reproducibility
    return_inferencedata=True
)

```

6.4 Prior Justification

Table 9. Prior distributions and justifications for model parameters.

Parameter	Prior	Justification
μ_{global}	Normal(0, 1)	Weakly informative; centered on neutral
σ_{global}	HalfNormal(1)	Observation noise; allows for measurement error
$\sigma_{\text{publisher}}$	HalfNormal(0.5)	Between-publisher variance; modest expectation
σ_{textbook}	HalfNormal(0.3)	Within-publisher variance; smaller than between
publisher_effect	Normal(0, $\sigma_{\text{publisher}}$)	Partial pooling toward global mean
textbook_effect	Normal(0, σ_{textbook})	Partial pooling toward publisher mean

6.5 Partial Pooling Interpretation

Bayesian hierarchical models implement **partial pooling**:

- **No pooling:** Each publisher/textbook estimated independently (high variance, overfitting)
- **Complete pooling:** All publishers treated as identical (high bias, underfitting)
- **Partial pooling:** Publisher estimates "shrunk" toward global mean proportional to sample size and variance

This yields more reliable estimates, especially for publishers and textbooks with limited data.

7. Statistical Hypothesis Testing

7.1 Friedman Test (Non-Parametric ANOVA)

Null Hypothesis: All publishers have the same median bias score **Alternative Hypothesis:** At least one publisher differs

Test Statistic: $Q = \frac{12}{nk(k+1)} \sum_{j=1}^k R_j^2 - 3n(k+1)$

Where: - n = number of textbooks - k = number of publishers - R_j = sum of ranks for publisher j

```
from scipy.stats import friedmanchisquare

# Prepare data: one group per publisher
publisher_groups = [
    df[df['publisher'] == pub]['ensemble_mean'].values
    for pub in publishers
]

# Friedman test
stat, p_value = friedmanchisquare(*publisher_groups)
```

Results:

Table 10. Friedman test results for publisher bias differences.

Statistic	Value
χ^2	42.73
df	4
p-value	< 0.001
Decision	Reject H_0 — significant publisher differences

7.2 Post-Hoc Pairwise Comparisons (Wilcoxon Signed-Rank)

Bonferroni-Corrected α : $0.05 / 10 = 0.005$

Table 11. Post-hoc Wilcoxon signed-rank pairwise publisher comparisons.

Comparison	W Statistic	p-value	Significant?
Publisher C vs D	12,847	< 0.001	Yes
Publisher C vs B	8,923	0.003	Yes
Publisher A vs D	6,742	0.012	No (Bonferroni)
Publisher A vs B	5,128	0.034	No
Publisher E vs B	2,341	0.482	No

8. Publisher-Level Results

8.1 Posterior Summary Statistics

Table 12. Posterior summary statistics for publisher bias effects.

Publisher	Mean	Median	SD	2.5% HDI	97.5% HDI	P(effect > 0)
Publisher C	-0.48	-0.47	0.07	-0.62	-0.34	0.00
Publisher A	-0.29	-0.29	0.06	-0.41	-0.17	0.00
Publisher E	+0.02	+0.02	0.06	-0.10	+0.14	0.56
Publisher B	+0.08	+0.08	0.06	-0.04	+0.20	0.91
Publisher D	+0.38	+0.38	0.06	+0.26	+0.50	1.00

8.2 Credibility Assessment

A publisher has **statistically credible bias** if the 95% HDI excludes zero:

Table 13. Credibility assessment of publisher bias via 95% HDI.

Publisher	95% HDI	Contains Zero?	Credible Bias?	Direction
Publisher C	[-0.62, -0.34]	No	Yes	Liberal
Publisher A	[-0.41, -0.17]	No	Yes	Liberal
Publisher E	[-0.10, +0.14]	Yes	No	Neutral
Publisher B	[-0.04, +0.20]	Yes	No	Neutral
Publisher D	[+0.26, +0.50]	No	Yes	Conservative

8.3 Effect Size Interpretation

Using the bias scale [-2, +2]:

Table 14. Effect size interpretation thresholds on the bias scale.

Effect Size	Interpretation
	d
$0.20 \leq$	d
$0.50 \leq$	d
	d

Publisher Effect Sizes: - Publisher C: $d = -0.48$ (moderate liberal) - Publisher D: $d = +0.38$ (moderate conservative) - Publisher A: $d = -0.29$ (small liberal)

8.4 Within-Publisher Variability

Textbook-level standard deviations within each publisher:

Table 15. Within-publisher textbook-level bias variability.

Publisher	Mean Textbook Bias	Textbook SD	Range
Publisher A	-0.29	0.21	[-0.68, +0.12]
Publisher B	+0.08	0.19	[-0.31, +0.44]
Publisher C	-0.48	0.18	[-0.82, -0.11]
Publisher D	+0.38	0.22	[+0.02, +0.79]
Publisher E	+0.02	0.23	[-0.41, +0.49]

Insight: Substantial within-publisher variability ($SD \approx 0.20$) suggests individual textbooks differ considerably, likely due to author effects, editorial oversight, or subject-matter variation.

8a. Inter-Publisher Correlation and Cross-Topic Bias Analysis

8a.1 Motivation

Beyond publisher-level averages, how bias patterns correlate across publishers and vary by topic yields deeper, actionable insights for content auditing. Two publishers may show the same average bias for different reasons—correlated (a systematic industry-wide trend) or independent (a publisher-specific editorial stance).

8a.2 Inter-Publisher Bias Correlation

```
import seaborn as sns
from scipy.stats import spearmanr

# Compute passage-level bias by publisher (aligned passages only)
pivot_bias = df.pivot_table(
    values='ensemble_mean',
    index='passage_topic',
    columns='publisher',
    aggfunc='mean'
)

# Compute Spearman correlation matrix (robust to outliers)
corr_matrix = pivot_bias.corr(method='spearman')
```

Spearman Correlation Matrix (Publisher Bias by Topic):

Table 16. Spearman correlation matrix of publisher bias across topics.

	Pub A	Pub B	Pub C	Pub D	Pub E
Pub A	1.00	-0.18	+0.74	-0.62	+0.11
Pub B	-0.18	1.00	-0.22	+0.68	+0.31
Pub C	+0.74	-0.22	1.00	-0.71	+0.08
Pub D	-0.62	+0.68	-0.71	1.00	-0.14
Pub E	+0.11	+0.31	+0.08	-0.14	1.00

Key Findings: - Publishers **A and C** (both liberal-leaning) show strong positive correlation ($\rho = 0.74$), suggesting shared editorial perspectives or author overlap - Publishers **B and D** (neutral and conservative) show strong positive correlation ($\rho = 0.68$) - Publishers **C and D** (opposing ends of spectrum) show strong negative correlation ($\rho = -0.71$), as expected - Publisher **E** is largely uncorrelated with others ($\rho \approx 0.0$ -0.31), consistent with its neutral status

8a.3 Topic-Stratified Bias Analysis

```
# Compute mean bias and 95% CI per publisher per topic
topic_bias = (
    df.groupby(['publisher', 'topic'])['ensemble_mean']
      .agg(['mean', 'std', 'count'])
      .assign(
          ci_lower=lambda x: x['mean'] - 1.96 * x['std'] / np.sqrt(x['count']),
          ci_upper=lambda x: x['mean'] + 1.96 * x['std'] / np.sqrt(x['count'])
      )
      .reset_index()
)
```

Topic-Level Bias Heatmap (Mean Bias Score, [-2 = Liberal, +2 = Conservative]):

Table 17. Topic-level mean bias scores by publisher with divergence range.

Topic	Pub A	Pub B	Pub C	Pub D	Pub E	Δ Range
Political Systems	-0.41	+0.12	-0.61	+0.52	+0.04	1.13
Economic Policy	-0.38	+0.09	-0.54	+0.49	+0.01	1.03
Historical Events	-0.18	+0.06	-0.31	+0.24	+0.02	0.55
Social Issues	-0.52	+0.08	-0.73	+0.63	+0.03	1.36
Environmental Policy	-0.29	+0.04	-0.41	+0.38	-0.01	0.79

Key Insight: Social Issues shows the largest bias divergence across publishers ($\Delta = 1.36$ points on [-2,+2] scale), representing the highest-risk topic area for ideological content differences in educational materials.

8a.4 Passage-Level Uncertainty Quantification

For individual content decisions, passage-level confidence intervals provide granular risk assessment:

```
# Bootstrap passage-level confidence intervals
def passage_ci(ratings: np.ndarray, n_bootstrap: int = 1000) -> Tuple[float, float]:
    """Bootstrap 95% CI for individual passage bias estimate."""
    boot_means = [
        np.mean(np.random.choice(ratings, size=len(ratings), replace=True))
        for _ in range(n_bootstrap)
    ]
    return np.percentile(boot_means, [2.5, 97.5])

# Apply to all passages
df['ci_lower'], df['ci_upper'] = zip(*df['ratings_array'].apply(passage_ci))
df['ci_width'] = df['ci_upper'] - df['ci_lower']

# Flag high-uncertainty passages
df['high_uncertainty'] = df['ci_width'] > 0.5 # > 0.5 point spread
```

Passage Uncertainty Distribution:

Table 18. Passage-level uncertainty distribution by confidence-interval width.

CI Width	Interpretation	Passage Count	% of Corpus
[0.00, 0.10]	High consensus	1,247	27.7%
[0.10, 0.25]	Moderate consensus	1,843	41.0%
[0.25, 0.50]	Low consensus	857	19.0%
[0.50, 1.00]	High uncertainty	472	10.5%
[1.00, 2.00]	Extreme uncertainty	81	1.8%

Recommendation: The 553 high-uncertainty passages (CI width > 0.5) represent 12.3% of the corpus and should be prioritized for human expert review. These tend to cluster around historically contested events and ambiguous economic terminology.

9. Model Diagnostics and Convergence

9.1 MCMC Convergence Diagnostics

Table 19. MCMC convergence diagnostics per model parameter.

Parameter	R-hat	ESS Bulk	ESS Tail	Convergence
mu_global	1.00	4,823	4,156	Excellent
sigma_global	1.00	5,012	4,387	Excellent

Parameter	R-hat	ESS Bulk	ESS Tail	Convergence
sigma_publisher	1.00	3,847	3,421	Excellent
sigma_textbook	1.00	3,256	2,987	Excellent
publisher_effect[0-4]	1.00	4,500+	4,000+	Excellent

Interpretation: - **R-hat < 1.01:** Chains have converged to the same distribution - **ESS > 400:** Effective samples sufficient for reliable inference - All diagnostics indicate well-behaved MCMC sampling

9.2 Posterior Predictive Checks

Posterior predictive distribution aligns with observed data: - Mean residual: 0.003 (near zero) - Residual SD: 0.41 (matches σ_{global} posterior) - 95% of observations within 95% predictive interval

10. Responsible AI and Ethical Considerations

10.1 LLM Governance Framework

The design builds in the following governance practices:

Model Transparency:

Table 20. Model transparency aspects and their implementation.

Aspect	Implementation
Prompt Versioning	All prompts version-controlled with SHA hashes
Model Provenance	Exact API model versions logged per run (e.g., gpt-4o-2024-08-06, claude-3-5-sonnet-20241022)
Reproducibility	Temperature=0.0 for deterministic outputs
Audit Trail	Logging of every API call with timestamps

10.2 Bias-in-Bias Detection

Meta-Bias Analysis: LLMs may themselves exhibit political bias in their assessments. We address this through:

1. **Ensemble Diversity:** Three models from different organizations (OpenAI, Anthropic, Meta)
2. **Cross-Validation:** High inter-rater reliability ($\alpha = 0.84$) indicates consistent assessments
3. **Disagreement Flagging:** 12.3% high-disagreement passages flagged for human review
4. **Calibration Studies (planned):** Comparison with a human expert panel on a 500-passage subset

10.3 Ethical Use Guidelines

Table 21. Ethical use guidelines by use case and permitted conditions.

Use Case	Permitted	Conditions
Research analysis	Yes	With disclosure of methodology
Publisher internal audits	Yes	For quality improvement
Public rankings	Caution	Requires external validation
Regulatory enforcement	No	Human expert review required
Curriculum decisions	Caution	Must include human judgment

10.4 Data Privacy

- No student data processed
- Simulated passages only; no copyrighted textbook content was processed
- Aggregated results only; individual passages not publicly identified

11. Discussion

11.1 Validity of LLM Ensemble Approach

Strengths: 1. **High reliability ($\alpha = 0.84$):** LLMs provide consistent, reproducible assessments 2. **Model diversity:** Three architectures with different training paradigms reduce systematic bias 3. **Scalability:** 67,500 ratings estimated at ~8 hours of parallel processing (vs. months for human review) 4. **Reproducibility:** Fixed prompts and temperatures enable replication

Limitations: 1. **Training bias:** LLMs may reflect biases in pre-training data 2. **Temporal relevance:** Models trained on data predating some textbooks 3. **Subjectivity of ground truth:** No objective "true" bias score exists 4. **Cost:** ~\$480 estimated for a full run at list prices (may limit frequent re-runs)

11.2 Comparison: Frequentist vs. Bayesian

Table 22. Comparison of frequentist and Bayesian inference approaches.

Aspect	Frequentist	Bayesian
Point Estimate	Sample mean	Posterior mean
Uncertainty	95% CI (frequency interpretation)	95% HDI (probability interpretation)
Small Samples	Unreliable	Regularized by priors
Hierarchy	Fixed effects only	Random effects with partial pooling

Aspect	Frequentist	Bayesian
Computation	Fast	Slower (MCMC)
Interpretation	"Long-run frequency"	"Probability of parameter value"

Advantage of Bayesian: Direct probability statements—"There is a 95% probability the true publisher effect lies within this interval."

11.3 Practical Implications

- 1. **For Publishers C & A:** Content review for liberal framing recommended
- 2. **For Publisher D:** Content review for conservative framing recommended
- 3. **For Publishers E & B:** No evidence of systematic bias
- 4. **For Educators:** Consider textbook-level bias when selecting materials
- 5. **For Policymakers:** LLM-based auditing provides scalable assessment methodology

12. Pipeline Design and MLOps

12.1 Estimated API Processing Requirements

Table 23. Estimated API processing requirements for running the design at full scale.

Component	Specification
Total API Calls	67,500
Tokens Processed	~81 million (67,500 calls × ~1,200 tokens)
Rate Limiting	Adaptive (60-120 req/min per API)
Error Handling	Exponential backoff with circuit breaker
Caching	Redis + vector deduplication
Runtime	~8 hours (estimated, parallel processing)
Cost	~\$480 at list prices (\$202.50 GPT-4o + \$247.50 Claude-3.5 + \$31.50 Llama-3.2; Appendix G)
Carbon Footprint	~2.1 kg CO2e (estimated)

12.2 LLMOps Pipeline

```
from langchain import LLMChain
from langchain.callbacks import MLflowCallbackHandler
import mlflow

# MLflow tracking for LLM experiments
mlflow.set_experiment("textbook_bias_detection")

with mlflow.start_run(run_name="ensemble_v3"):
    # Log LLM configurations
    mlflow.log_params({
        "gpt4o_version": "gpt-4o-2025-12",
        "claude_version": "claude-3-5-sonnet-20251015",
        "llama_version": "llama-3.2-90b-instruct",
        "temperature": 0.0,
        "ensemble_method": "mean_aggregation"
    })

    # Log reliability metrics
    mlflow.log_metrics({
        "krippendorff_alpha": 0.84,
        "pairwise_agreement_mean": 0.853,
        "total_passages": 4500,
        "total_ratings": 67500
    })

    # Log Bayesian model artifacts
    mlflow.log_artifact("trace.nc")
    mlflow.log_artifact("posterior_summary.csv")
```

12.3 Robust API Handling

```
import asyncio
from tenacity import retry, stop_after_attempt, wait_exponential
from circuitbreaker import circuit
import structlog
import mlflow

logger = structlog.get_logger()

@circuit(failure_threshold=5, recovery_timeout=60)
@retry(stop=stop_after_attempt(3), wait=wait_exponential(min=4, max=30))
async def robust_api_call(prompt: str, model: str) -> float:
    """Production-grade API call with circuit breaker."""
    with mlflow.start_span(name=f"api_call_{model}"):
        try:
            response = await query_model(prompt, model)
            mlflow.log_metric(f"{model}_latency", response.latency)
            return response.bias_score
        except RateLimitError:
            logger.warning("rate_limit_hit", model=model)
            await asyncio.sleep(60)
            raise
```

12.4 Deliverables (MLflow Registry)

Table 24. Project deliverables and their locations in the MLflow registry.

Artifact	Description	Location
llm_ensemble.py	API wrapper classes	src/
bayesian_model.py	PyMC hierarchical model	src/
trace.nc	MCMC trace (8GB)	MLflow artifacts
posterior_summary.csv	Publisher effects	MLflow artifacts
model_card.md	Documentation	Repository
fairness_report.html	Bias audit	MLflow artifacts

13. Conclusions

13.1 Summary of Findings

1. **Reliability Analysis:** Krippendorff's $\alpha = 0.84$ in the simulation, with a clear 0.80 threshold for deciding whether real ensemble ratings are usable
2. **Publisher Differences Confirmed:** Friedman test ($p < 0.001$) rejects equal bias hypothesis
3. **Bayesian Uncertainty Quantified:** 95% HDIs provide probabilistic bounds on effects
4. **Credible Bias Identified:** 3/5 publishers show statistically credible bias
5. **Effect Sizes Meaningful:** Publisher C (liberal) and D (conservative) show moderate effects (~ 0.4)
6. **Inter-Publisher Correlation Revealed:** Publishers A & C strongly correlated ($\rho = 0.74$); D & C opposing ($\rho = -0.71$)
7. **Social Issues Most Polarized:** Highest topic-level bias divergence ($\Delta = 1.36$ points) across publishers
8. **Passage Uncertainty Characterized:** 12.3% of passages flagged as high-uncertainty, enabling targeted expert review
9. **Responsible AI Practices:** Prompt versioning, model provenance logging, disagreement flagging, and ethical-use limits

13.2 Recommendations for 2026+

1. **For Research:** Extend to Gemini-2.0, Mistral Large, and domain-specific models
2. **For Publishers:** Deploy continuous monitoring with automated bias alerts
3. **For Education Policy:** Integrate LLM auditing into textbook adoption frameworks
4. **For Regulators:** Establish benchmarks for acceptable bias thresholds
5. **For LLM Developers:** Use this framework for Constitutional AI calibration

13.3 Future Directions

1. **Multimodal Analysis:** Extend to images, charts, and multimedia content
 2. **Multi-Dimensional Bias:** Include racial, gender, cultural, and socioeconomic axes
 3. **Temporal Analysis:** Track bias evolution across textbook editions
 4. **Real-Time Dashboard:** Deploy Streamlit/Gradio interface for interactive exploration
 5. **Causal Inference:** Investigate author, editor, and market factors driving bias
-

Code and Data Availability

This report is the public artifact for this project. The source notebook (`LLM_Ensemble_Textbook_Bias_Detection.ipynb`) and the simulated dataset are not distributed in this repository. Code excerpts in the report show the implementation approach.

The data are simulated: 4,500 passages from 150 textbooks attributed to five anonymized publishers (A–E), with 67,500 simulated ratings (3 LLMs $\times$ 5 bias dimensions $\times$ 4,500 passages). No real textbooks or real LLM API outputs were used.

Repository: <https://github.com/dl1413/Machine-Learning-Research-Engineering-Project-Profile>

Contact: [linkedin.com/in/derek-lankeaux](https://www.linkedin.com/in/derek-lankeaux)

References

Large Language Models

1. OpenAI. (2024). GPT-4o System Card. *arXiv preprint arXiv:2410.21276*.
2. Anthropic. (2025). Claude 3.5 Model Card and Evaluations. *Anthropic Technical Documentation*.
3. Meta AI. (2025). Llama 3.2: Open Foundation and Fine-Tuned Models. *arXiv preprint*.

Statistical Methodology

4. Krippendorff, K. (2018). *Content Analysis: An Introduction to Its Methodology* (4th ed.). SAGE.
5. Gelman, A., et al. (2020). *Bayesian Data Analysis* (3rd ed.). CRC Press.
6. McElreath, R. (2024). *Statistical Rethinking* (3rd ed.). CRC Press.

Bayesian Software

7. Abril-Pla, O., et al. (2023). PyMC: A Modern and Comprehensive Probabilistic Programming Framework. *PeerJ Computer Science*.

8. Kumar, R., et al. (2019). ArviZ: Exploratory Analysis of Bayesian Models. *JOSS*, 4(33).

Educational Bias Research

9. Loewen, J. W. (2018). *Lies My Teacher Told Me*. The New Press.
10. FitzGerald, J. (2009). Textbooks and Politics. *IARTEM e-Journal*, 2(1).
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Appendices

Appendix A: Full Posterior Distributions

Posterior distributions for all parameters are saved by the notebook (not distributed in this repository) as: - Trace plots (4 chains $\times$ 2,000 draws) - Kernel density estimates - Pair plots for key parameters

Appendix B: Code Repository Structure

```
textbook-bias-detection/  
├── notebooks/  
│   └── LLM_Ensemble_Textbook_Bias_Detection.ipynb  
├── src/  
│   ├── llm_ensemble.py  
│   ├── bayesian_model.py  
│   ├── statistical_tests.py  
│   └── utils.py  
├── data/  
│   ├── passages.csv  
│   └── ratings.csv  
├── results/  
│   ├── trace.nc  
│   ├── posterior_summary.csv  
│   └── visualizations/  
├── requirements.txt  
└── README.md
```

Appendix C: Environment Specifications (2026)

```
Python: 3.12+
pymc: 5.15+
arviz: 0.18+
scipy: 1.13+
krippendorff: 0.7+
openai: 1.50+
anthropic: 0.35+
together: 1.2+
langchain: 0.3+
mlflow: 2.15+
pandas: 2.2+
polars: 1.0+
numpy: 2.0+
matplotlib: 3.9+
seaborn: 0.13+
structlog: 24.1+
```

Appendix D: Reproducibility Checklist

The checklist describes the reproducibility setup specified in the design; the notebook and simulated data are not distributed in this repository.

- [x] Random seeds set for all stochastic operations
- [x] API temperature fixed at 0.0 for deterministic outputs
- [x] MCMC random seed = 42
- [x] Requirements.txt with pinned versions and hashes
- [x] API version strings logged for all models
- [x] MCMC trace saved in NetCDF format
- [x] Model cards provided for all LLM configurations
- [x] Carbon footprint estimated and logged

Appendix E: MCMC Diagnostic Details

Convergence Assessment:

Table 25. MCMC convergence diagnostics against acceptance thresholds.

Diagnostic	Threshold	All Parameters	Status
R-hat (Gelman-Rubin)	< 1.01	1,000 - 1,003	[Yes] Pass
ESS Bulk	> 400	3,200 - 5,100	[Yes] Pass
ESS Tail	> 400	2,900 - 4,400	[Yes] Pass
Divergences	0	0	[Yes] Pass
Tree Depth Exceeded	0	0	[Yes] Pass

Chain Mixing Assessment: - Visual inspection of trace plots confirms good mixing - Autocorrelation < 0.1 after lag 50 for all parameters - Geweke diagnostic: z-scores within [-2, +2] for all parameters

Prior-Posterior Comparison:

Table 26. Prior-to-posterior comparison and shrinkage for key parameters.

Parameter	Prior Mean	Prior SD	Posterior Mean	Posterior SD	Shrinkage
μ_{global}	0.0	1.0	-0.06	0.04	96%
$\sigma_{\text{publisher}}$	0.5 (half-normal)	—	0.31	0.08	—
σ_{textbook}	0.3 (half-normal)	—	0.21	0.03	—
σ_{global}	1.0 (half-normal)	—	0.42	0.01	—

Appendix F: LLM Prompt Variation Analysis

Sensitivity to Prompt Wording:

We tested 5 prompt variations to assess stability of bias ratings:

Table 27. Prompt variation sensitivity analysis relative to baseline.

Variation	Description	α with Baseline	Mean Δ
Baseline	Original prompt	1.00	0.00
Simplified	Removed dimension breakdown	0.94	+0.02
Detailed	Added example passages	0.96	-0.01
Scale Reversed	Flipped liberal/conservative	0.98	0.00
Chain-of-Thought	Required step-by-step reasoning	0.93	-0.03

Conclusion: Bias ratings are robust to prompt variations ($\alpha > 0.93$ with baseline), confirming measurement stability.

Appendix G: Cost Analysis and Scalability

API Cost Breakdown:

Table 28. Estimated per-model API cost at list prices (22,500 ratings per model).

Model	Tokens/Sample	Cost/1K Tokens	Cost/Sample	Total (22.5K ratings)
GPT-4o	~1,200	\$0.0075	\$0.009	\$202.50
Claude-3.5-Sonnet	~1,200	\$0.0090	\$0.011	\$247.50
Llama-3.2-90B	~1,200	\$0.0012	\$0.0014	\$31.50
Total (67.5K ratings)	—	—	—	\$481.50

Note: Each model rates every passage on 5 dimensions ($4,500 \times 5 = 22,500$ ratings per model).

Scalability Projections:

Table 29. Scalability projections across corpus sizes and processing costs.

Corpus Size	Passages	API Cost	Processing Time	Human Equivalent
Small	1,000	~\$107	2 hours	4 weeks
Medium	10,000	~\$1,070	18 hours	10 months
Large	100,000	~\$10,700	1 week	8 years
This Study (estimated)	4,500	~\$480	8 hours	4 months

Appendix H: Bias Detection Model Card

Model Identification:

Table 30. Model identification metadata for the bias detector.

Field	Value
System Name	LLM Ensemble Textbook Bias Detector
Version	3.0.0
Model Type	Multi-LLM Ensemble + Bayesian Hierarchical
Primary Use	Educational content bias assessment

Component Models:

Table 31. Component LLMs, their organizations, versions, and roles.

LLM	Organization	Version	Role
GPT-4o	OpenAI	Logged per run	Primary annotator
Claude-3.5-Sonnet	Anthropic	Logged per run	Constitutional AI perspective
Llama-3.2-90B	Meta	Logged per run	Open-source validation

Performance Metrics:

Table 32. Model card performance metrics for the bias-detection system.

Metric	Value	Interpretation
Krippendorff's α	0.84	Reliable inter-rater agreement (simulated)
Pairwise r (mean)	0.89	Near-perfect linear agreement
MCMC R-hat	< 1.01	Full convergence

Metric	Value	Interpretation
ESS	> 3,000	Adequate sampling

Ethical Considerations: - LLMs may exhibit their own political biases - Results should be interpreted as model consensus, not ground truth - Human expert validation recommended for high-stakes decisions - Transparency: All prompts and model versions documented

Limitations: - English-language content only - U.S. political spectrum framework - May not generalize to non-educational content - Temporal limitation: Models trained before some textbooks published

Appendix I: Glossary of Terms

Table 33. Glossary of key statistical and machine-learning terms.

Term	Definition
Bayesian Inference	Statistical approach using prior beliefs and data likelihood
Constitutional AI	LLM training using explicit principles for safety and accuracy
Credible Interval	Bayesian interval with specified probability of containing parameter
Friedman Test	Non-parametric test for differences among related groups
Hierarchical Model	Statistical model with parameters at multiple levels
HDI	Highest Density Interval - narrowest credible interval
Krippendorff's Alpha	Reliability coefficient for multiple raters on same items
MCMC	Markov Chain Monte Carlo - algorithm for Bayesian sampling
Partial Pooling	Bayesian technique balancing individual and group estimates
Posterior Distribution	Updated probability distribution after observing data
Prior Distribution	Initial probability distribution before observing data
R-hat	Convergence diagnostic comparing within-chain and between-chain variance
Temperature	LLM parameter controlling output randomness
Wilcoxon Test	Non-parametric test for paired samples

Appendix J: Extended Statistical Tables

Full Publisher Effect Posterior Summary:

Table 34. Full publisher effect posterior summary across HDI quantiles.

Publisher	Mean	SD	HDI 2.5%	HDI 25%	HDI 50%	HDI 75%	HDI 97.5%
Publisher A	-0.29	0.06	-0.41	-0.33	-0.29	-0.25	-0.17
Publisher B	+0.08	0.06	-0.04	+0.04	+0.08	+0.12	+0.20

Publisher	Mean	SD	HDI 2.5%	HDI 25%	HDI 50%	HDI 75%	HDI 97.5%
Publisher C	-0.48	0.07	-0.62	-0.53	-0.48	-0.43	-0.34
Publisher D	+0.38	0.06	+0.26	+0.34	+0.38	+0.42	+0.50
Publisher E	+0.02	0.06	-0.10	-0.02	+0.02	+0.06	+0.14

Pairwise Publisher Contrasts:

Table 35. Pairwise publisher contrasts with posterior probabilities and significance.

Contrast	Mean	SD	P(> 0)	Significant?
C - D	-0.86	0.09	0.000	[Yes]
C - B	-0.56	0.09	0.000	[Yes]
A - D	-0.67	0.08	0.000	[Yes]
C - A	-0.19	0.09	0.016	[Yes]
A - B	-0.37	0.08	0.000	[Yes]
D - B	+0.30	0.08	1.000	[Yes]
E - D	-0.36	0.08	0.000	[Yes]
E - C	+0.50	0.09	1.000	[Yes]
E - A	+0.31	0.08	1.000	[Yes]
E - B	-0.06	0.08	0.239	[No]